MATHCOACH · GRADE 10 · CAPS / IEB / CAMBRIDGE

ALGEBRA ASSAULT

FIELD MANUAL

The companion handbook to the **Algebra Assault** game. Five missions. Five algebra weapons. **One Grade 10 you actually understand.**

YOUR FIVE WEAPONS

- Linear
- Quadratic
- Simultaneous
- Exponents
- Inequalities

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MISSION BRIEFING

Welcome, Operator.

You downloaded Algebra Assault to *play*. This manual is what makes you **win**.

Grade 10 algebra is the gate. Pass through it and Grade 11 and 12 stay open. Trip on it and the rest of high school maths gets brutal. This manual covers the five weapons you need:

YOUR FIVE WEAPONS

1. **Linear Equations** — the foundation. If this is shaky, everything wobbles.
2. **Quadratic Equations** — the boss-killer. Factorise or die.
3. **Simultaneous Equations** — two unknowns, one move. Substitution & elimination.
4. **Exponents** — the rules that bend reality. Negative powers, equations, simplifying.
5. **Inequalities** — the sign-flip trap. The exam-paper marks everyone drops.

How this manual is built

Every chapter follows the same loop. Short. Visual. Built for brains that bounce.

- **HOOK** Why this matters in 30 seconds.
- **TEACH** The core idea — no fluff, no padding.
- **EXAMPLES** 3–4 worked solutions, every step shown.
- **TRAPS** The mistakes the marker is waiting for.
- **DRILLS** Quick practice — get the reps in.
- **BOSS BATTLE** Exam-style questions. Marked-up solutions in the back.

PRO TIP — HOW TO ACTUALLY USE THIS

Don't read it like a novel. Open the chapter that hurts most right now. Do the worked examples *with* a pen. Then jump into the game's matching mission and put the weapon to work.

Curriculum coverage

This manual covers the algebra core that's the same across:

CAPS Grade 10 South African National Curriculum
IEB Independent Examinations Board, Grade 10
CAM Cambridge IGCSE 0580 / 0606 (matching algebra topics)

Where the three diverge — like IEB introducing sum/difference of cubes early — we flag it. Cambridge learners: a few topics here go slightly beyond your core. Treat them as a head-start.

PRE-FLIGHT CHECK

Before you load up the first mission...

Pick the option below that's actually true right now. **Be honest** — this is just for you. The same dial appears at the end of the manual so you can see how far you've moved.

HOW DO YOU FEEL ABOUT GRADE 10 ALGEBRA RIGHT NOW?

?

No clue

Letters and numbers in the same equation still feels like a trick.

~

Some bits stick

Linear is OK-ish. Quadratics and exponents start to wobble.

✓

Mostly there

I can do most of it. I just lose marks I shouldn't.

THE ONE TOPIC THAT SCARES YOU MOST

Write the topic above. Then open Mission 01 and don't close this manual until you've found that topic and worked the examples with a pen.

CONTENTS

Mission Index

The Algebra Assault Game — Mission Map

This manual mirrors the game. Each chapter teaches the algebra you'll use against a specific boss:

Manual Mission	Game Boss	Weapon Unlocked
01 Linear Equations	The Slope Sentinel	Inverse Operations Cannon
02 Quadratic Equations	The Parabola Beast	Factorisation Blade
03 Simultaneous Equations	The Twin Tyrants	Substitution & Elimination Dual-Wield
04 Exponents	The Power Warden	Index Law Gauntlet
05 Inequalities	The Sign-Flipper	Number Line Sniper

REAL TALK

Algebra is not magic. It's pattern recognition with rules. Once you see the pattern, you can't unsee it. This manual is here to make the patterns obvious.

MISSION 01

Linear Equations

THE FOUNDATION · Boss: The Slope Sentinel

The Hook

A linear equation is any equation where the variable is to the power of 1 — no squares, no roots, just x.

Linear equations are how you answer questions like: *How many hours of tutoring before I hit 70%? How much airtime can I buy with R85? When will two cars meet?*

Master this and you've cracked maybe 30% of every test you'll write this year. Mess this up and quadratics, simultaneous, and inequalities all collapse — because they're all built on linear moves.

The Core Idea — "Whatever you do to one side, do to the other"

An equation is a balance. The $=$ sign is the pivot. To find x, you peel away everything attached to it using the **inverse operation**:

If you see...	Undo with...
$+5$	-5 on both sides
-7	$+7$ on both sides
$\times 3$	$\div 3$ on both sides
$\div 4$	$\times 4$ on both sides

GOAL OF EVERY LINEAR EQUATION

Get x alone on one side. That's it. That's the whole game.

TRAP #1 — "CHANGE SIDE, CHANGE SIGN" WITHOUT THINKING

Most learners chant "*change side change sign*" and bypass understanding. That works… until brackets, fractions, or unknowns on both sides show up. Then they break.

Fix: Always think *inverse operation*, not *change sides*. Same answer, no surprises.

WORKED EXAMPLES · MISSION 01

EXAMPLE 1 · Basic Linear

Solve for x: $3x + 5 = 17$

1. **Subtract 5** from both sides to isolate the $3x$ term.

   $3x + 5 - 5 = 17 - 5$

   $3x = 12$

2. **Divide both sides by 3** to free x.

   $x = 12 \div 3$

   $\mathbf{x = 4}$

   Check: 3(4) + 5 = 12 + 5 = 17 ✓

EXAMPLE 2 · With Brackets

Solve for x: $2(x - 3) = x + 4$

1. **Distribute** the 2 across the bracket on the left.

   $2x - 6 = x + 4$

2. **Move x terms to the left** by subtracting x from both sides.

   $2x - x - 6 = x - x + 4$

   $x - 6 = 4$

3. **Add 6** to both sides.

   $\mathbf{x = 10}$

   Check: 2(10 − 3) = 14 and 10 + 4 = 14 ✓

EXAMPLE 3 · With Fractions

Solve for x: $x/3 + 2 = x/2 - 1$

1. **Kill the fractions first.** The LCM of 3 and 2 is 6. Multiply every term by 6.

   $6(x/3) + 6(2) = 6(x/2) - 6(1)$

   $2x + 12 = 3x - 6$

2. **Move x terms to one side** (subtract 2x from both).

   $12 = x - 6$

3. **Add 6.**

   $\mathbf{x = 18}$

   Check: 18/3 + 2 = 6 + 2 = 8 and 18/2 − 1 = 9 − 1 = 8 ✓

EXAMPLE 4 · Literal Equation (rearranging a formula)

Solve for r: $C = 2\pi r$ (the circumference formula)

1. We want r alone. Currently it's multiplied by 2π.
2. **Divide both sides by 2π.**

$C \div 2\pi = r$

$\mathbf{r = C / (2\pi)}$

Same rule — inverse operation. Just doing it with letters.

TRAP #2 — DISTRIBUTING THE NEGATIVE WRONG

If you have $-2(x - 3) = -2x + 6$ (both signs flip). Most learners write $-2x - 6$ and lose easy marks.

DRILLS · MISSION 01 · SOLVE FOR X

1. $4x - 7 = 21$
2. $5x + 3 = 2x + 18$
3. $3(x + 4) = 2x + 15$
4. $2(x - 1) = 3(x - 4)$
5. $x/4 + 1 = 5$
6. $(2x + 1)/3 = 5$
7. $x/2 + x/3 = 10$
8. Solve for h: $A = (1/2)bh$

PRO TIP — WHEN IN DOUBT, PLUG IT BACK IN

If you're not sure your answer is right, substitute it into the original equation. If both sides match, you nailed it. If they don't, the error is upstream.

⚔ BOSS BATTLE 01 · THE SLOPE SENTINEL

Exam-style. Time yourself: 10 minutes. Solutions on p. 44.

1. Solve for x: $5(2x - 3) - 4(x - 1) = 9$
2. Solve for x: $(x - 2)/3 - (x + 1)/4 = 1$
3. The perimeter of a rectangle is 46 cm. The length is 5 cm more than the width. Find the dimensions. (Set up an equation, then solve.)
4. Make v the subject: $s = ut + (1/2)at^2$ [treat as linear in v after rearranging]
5. Solve for x: $3(x - 2) - 2(2x + 1) = -x - 8$ [careful with the negatives]

MISSION 01 SELF-CHECK — RATE YOUR CONFIDENCE

☐ Still confused ☐ Getting there ☐ Solid

GAME LINK

Open Algebra Assault → Mission 01: The Slope Sentinel. Use the Inverse Operations Cannon. Aim for a 5-streak before moving to Mission 02.

MISSION 02

Quadratic Equations

THE BOSS-KILLER · Boss: The Parabola Beast

The Hook

A quadratic equation has x^2 in it. That single squared term changes the rules: a quadratic can have **two answers**, one answer, or no real answer at all.

In Grade 10 we solve quadratics **by factorisation**. The quadratic formula and completing the square arrive in Grade 11. Don't skip ahead — Grade 10 factorisation is the skill the markers care about.

The Core Idea — The Zero Product Principle

If two things multiply to give 0, then **at least one of them must be 0**.

$$\text{If } \ A \times B = 0 \ \text{ then } \ A = 0 \ \text{ OR } \ B = 0$$

That's the whole engine. You factorise the quadratic into two brackets multiplying to give zero, then each bracket gives you one solution.

THE 4-STEP QUADRATIC PROTOCOL

1. **Standard form** — get everything on one side: $ax^2 + bx + c = 0$
2. **Common factor** — pull out anything shared (don't skip this — it makes the trinomial simpler)
3. **Factorise** — into two brackets
4. **Set each bracket = 0** and solve

TRAP #1 — THE "MISSED SOLUTION" DISASTER

If you solve $x^2 = 25$ and write $x = 5$, you just lost half the marks. **x = 5 OR x = −5**. Both 5^2 and $(-5)^2$ equal 25.

TRAP #2 — SKIPPING STANDARD FORM

You cannot factorise $x^2 + 3x = 10$ directly. You must first move everything to one side: $x^2 + 3x - 10 = 0$. Then factorise. Standard form is non-negotiable.

TRAP #3 — DIVIDING BOTH SIDES BY X

Given $x^2 = 4x$, do NOT divide both sides by x. You'll lose the solution $x = 0$. Always move everything to one side and factor out x.

WORKED EXAMPLES · MISSION 02

EXAMPLE 1 · Classic Trinomial

Solve: $x^2 - 5x + 6 = 0$

1. Already in standard form. ✓
2. **Factorise the trinomial.** Find two numbers that multiply to give **+6** and add to **−5**. → −2 and −3.

   $(x - 2)(x - 3) = 0$
3. **Zero product principle.**

   $x - 2 = 0 \quad \text{OR} \quad x - 3 = 0$

   $\mathbf{x = 2 \quad OR \quad x = 3}$

EXAMPLE 2 · Common Factor First

Solve: $2x^2 - 8x = 0$

1. Standard form ✓ (right side is 0).
2. **Common factor:** both terms share $2x$.

   $2x(x - 4) = 0$
3. **Each factor = 0.**

   $2x = 0 \quad \text{OR} \quad x - 4 = 0$

   $\mathbf{x = 0 \quad OR \quad x = 4}$

   If you'd "divided by x" at step 2, you'd have missed x = 0 entirely. That's the trap.

EXAMPLE 3 · Difference of Squares

Solve: $x^2 - 16 = 0$

1. Standard form ✓.
2. **Difference of squares pattern:** $a^2 - b^2 = (a - b)(a + b)$.

   $(x - 4)(x + 4) = 0$
3. **Zero product.**

   $\mathbf{x = 4 \quad OR \quad x = -4}$

EXAMPLE 4 · Not in Standard Form

Solve: $x^2 = 4x + 12$

1. **Get to standard form.** Move everything to the left.

   $x^2 - 4x - 12 = 0$

2. **Factorise.** Two numbers: multiply to −12, add to −4. → −6 and +2.

   $(x - 6)(x + 2) = 0$

3. **Solve each.**

   $\mathbf{x = 6 \quad OR \quad x = -2}$

   Check: $6^2 = 36$ *and* $4(6) + 12 = 36$ ✓ | $(-2)^2 = 4$ *and* $4(-2) + 12 = 4$ ✓

PRO TIP — THE "MULTIPLY TO / ADD TO" TRICK

For $x^2 + bx + c$: find two numbers that **multiply to c** and **add to b**. Practice this until it's instant — it's the single most-used skill on every Grade 10 paper.

IEB EXTRA — SUM & DIFFERENCE OF CUBES (GRADE 10)

IEB Grade 10 introduces:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$
$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

CAPS only introduces these in Grade 12 via the factor theorem. Cambridge: covered in 0606 Additional Maths.

DRILLS · MISSION 02 · SOLVE

1. $x^2 - 7x + 12 = 0$
2. $x^2 + 2x - 15 = 0$
3. $x^2 - 9 = 0$
4. $3x^2 - 12x = 0$
5. $x^2 + 5x = 0$
6. $x^2 = 6x - 5$
7. $2x^2 + 7x + 3 = 0$
8. $(x + 1)(x - 4) = 6$ ⚠ standard form first!

PRO TIP — WHEN THE RIGHT SIDE ISN'T ZERO

Drill #8 catches everyone. $(x + 1)(x - 4) = 6$ — you CANNOT set each bracket to 6. Expand first, then move 6 across, THEN factorise the new trinomial.

⚔ BOSS BATTLE 02 · THE PARABOLA BEAST

Exam-style. Time yourself: 12 minutes. Solutions on p. 44.

1. Solve for x: $x^2 - 11x + 28 = 0$
2. Solve for x: $2x^2 - 5x - 3 = 0$
3. Solve for x: $x(x - 4) = 5(x - 4)$ [expand both sides first]
4. The product of two consecutive positive integers is 72. Find them. (Let the smaller = x.)
5. Solve for x: $(2x + 1)^2 = 25$ [two ways: expand, or square root carefully]

MISSION 02 SELF-CHECK

□ Still confused □ Getting there □ Solid

GAME LINK

Open Algebra Assault → Mission 02: The Parabola Beast. Use the Factorisation Blade. The Beast has TWO weak points — that's the two solutions. Hit both.

MISSION 03

Simultaneous Equations

TWO UNKNOWNS, ONE STRIKE · Boss: The Twin Tyrants

The Hook

One equation, one unknown. Two equations, two unknowns. The numbers go up, but the rules barely change.

Simultaneous equations answer "two things at once" problems: prices of two items, where two lines cross, ratios of mixtures, ages, speeds.

Two Methods. Pick One. Master Both.

METHOD A — SUBSTITUTION

1. Make one variable the subject in one equation
2. Substitute that expression into the other equation
3. Solve the now-single-variable equation
4. Back-substitute to find the second variable

Best when: one variable already has a coefficient of 1 (or is already alone).

METHOD B — ELIMINATION

1. Multiply one or both equations so a variable's coefficients match
2. Add or subtract the equations to cancel that variable
3. Solve the resulting single-variable equation
4. Back-substitute to find the second variable

Best when: coefficients are messy / both variables have coefficients > 1.

VISUAL INTUITION

Each linear equation is a line on the Cartesian plane. The simultaneous solution is the **point where the two lines cross**. One x, one y, one intersection.

TRAP — FORGETTING TO BACK-SUBSTITUTE

You solved for x. You wrote it down. You moved on. **Stop.** The question wants *both* x and y. Always substitute back to find the second variable.

TRAP — SIGN ERRORS WHEN SUBTRACTING EQUATIONS

When you subtract one equation from another, **every term changes sign**. Write it out — don't do it in your head. The sign of the constant gets dropped the most.

WORKED EXAMPLES · MISSION 03

EXAMPLE 1 · Substitution (clean)

Solve: $y = 2x + 1$ and $3x + y = 11$

1. y is already the subject in equation (1). **Substitute** into (2).

   $3x + (2x + 1) = 11$

2. **Simplify and solve for x.**

   $5x + 1 = 11 \rightarrow 5x = 10 \rightarrow x = 2$

3. **Back-substitute** $x = 2$ into equation (1).

   $y = 2(2) + 1 = 5$

   $\mathbf{x = 2, \quad y = 5}$

EXAMPLE 2 · Elimination

Solve: $2x + 3y = 12$ …(1) and $4x - y = 10$ …(2)

1. **Match coefficients.** Multiply (2) by 3 so the y coefficient becomes 3.

   $12x - 3y = 30$ …(2′)

2. **Add (1) + (2′)** — the 3y and −3y cancel.

   $14x = 42 \rightarrow x = 3$

3. **Back-substitute** $x = 3$ into (1).

   $2(3) + 3y = 12 \rightarrow 3y = 6 \rightarrow y = 2$

   $\mathbf{x = 3, \quad y = 2}$

EXAMPLE 3 · Word Problem

Question: Two burgers and three sodas cost R85. One burger and five sodas cost R75. Find the cost of each.

1. **Define variables.** Let b = burger, s = soda.
2. **Set up equations.**

$2b + 3s = 85$ ...(1)

$b + 5s = 75$ ...(2)

3. **Substitution:** from (2), b = 75 − 5s. Substitute into (1).

$2(75 - 5s) + 3s = 85$

$150 - 10s + 3s = 85$

$-7s = -65 \rightarrow s = R9{,}29$... wait, let's adjust.

Real-world prices should be clean — let's try cleaner numbers in practice. For this teaching example: s ≈ R9.29, b ≈ R28.57. In an exam, expect whole-rand answers.

EXAMPLE 4 · Both Methods, Cleaner Numbers

Solve: $x + 2y = 8$ ...(1) and $3x - y = 3$ ...(2)

Substitution route:

From (1): $x = 8 - 2y$

Into (2): $3(8 - 2y) - y = 3 \rightarrow 24 - 6y - y = 3 \rightarrow -7y = -21 \rightarrow y = 3$

Then $x = 8 - 2(3) = 2$

Elimination route:

Multiply (2) by 2: $6x - 2y = 6$

Add to (1): $7x = 14 \rightarrow x = 2$, then $y = 3$

Same answer: $x = 2, y = 3$

Both methods always reach the same answer. Pick whichever gets you there fastest.

DRILLS · MISSION 03 · SOLVE FOR X AND Y

1. $y = x + 3$; $2x + y = 12$
2. $x + y = 7$; $x - y = 1$
3. $3x + 2y = 16$; $x + y = 6$
4. $2x - y = 5$; $x + 2y = 5$
5. $4x + 3y = 17$; $2x - y = 1$
6. $5x + 2y = 11$; $3x - y = 8$
7. The sum of two numbers is 24. Their difference is 6. Find them.
8. 3 pens and 2 books cost R75. 1 pen and 4 books cost R85. Find each.

PRO TIP — ALWAYS LABEL EQUATIONS (1) AND (2)

When examiners can't follow your work, you lose method marks. Number your equations. Show which one you're multiplying. Show the new (1′) or (2′). It's clean and gets you the marks.

⚔ BOSS BATTLE 03 · THE TWIN TYRANTS

Exam-style. Time yourself: 15 minutes. Solutions on p. 45.

1. Solve: $2x + 5y = 16$; $3x - 2y = 5$
2. Solve: $x/2 + y/3 = 5$; $x - y = 4$ [clear fractions first]
3. The cost of 5 oranges and 3 apples is R36. 2 oranges and 4 apples cost R34. Find the cost of each.
4. A two-digit number has a sum of digits 11. When the digits are reversed, the number is 27 less than the original. Find the number. (Let tens = x, units = y; the number is $10x + y$.)
5. Solve graphically (sketch only): $y = 2x - 1$; $y = -x + 5$. State the intersection.

MISSION 03 SELF-CHECK

□ Still confused □ Getting there □ Solid

GAME LINK

Open Algebra Assault → Mission 03: The Twin Tyrants. The bosses come in pairs — pick the right method (substitution or elimination) based on what they're showing you. Wrong method = double damage.

MISSION 04

Exponents

THE REALITY BENDERS · Boss: The Power Warden

The Hook

Exponents are how we write very big numbers, very small numbers, growth, decay, compound interest, viruses, and most of physics. Every Grade 10 paper has exponent simplification AND an exponential equation.

The good news: there are only **six rules**. The bad news: learners confuse them constantly. This chapter locks them in.

The Six Exponent Laws

Rule	Example
$a^m \times a^n = a^{m+n}$ (same base, multiply → ADD powers)	$x^3 \times x^5 = x^8$
$a^m \div a^n = a^{m-n}$ (same base, divide → SUBTRACT powers)	$x^7 \div x^3 = x^4$
$(a^m)^n = a^{mn}$ (power of a power → MULTIPLY)	$(x^2)^3 = x^6$
$(ab)^n = a^n b^n$ (power distributes over multiplication)	$(2x)^3 = 8x^3$
$a^0 = 1$ (anything to the power 0 is 1 — except 0^0, undefined)	$7^0 = 1,\ (5x)^0 = 1$
$a^{-n} = 1 / a^n$ (negative power = reciprocal)	$2^{-3} = 1/8$

TRAP #1 — ADDING POWERS WHEN BASES ARE DIFFERENT

$2^3 \times 3^4$ does NOT equal 6^7. The "add powers" rule only works when the **bases are identical**.

TRAP #2 — DISTRIBUTING A POWER OVER ADDITION

$(a + b)^2$ is NOT $a^2 + b^2$. It's $a^2 + 2ab + b^2$. The exponent only distributes over *multiplication*, never addition.

TRAP #3 — CONFUSING $-A^2$ WITH $(-A)^2$

$-3^2 = -9$ (square first, then negate)
$(-3)^2 = 9$ (negate first, then square)

TRAP #4 — NEGATIVE EXPONENT ≠ NEGATIVE ANSWER

$2^{-3} = 1/8 = 0.125$. It's a small positive number, not −8. Negative powers flip the base into the denominator; they don't change the sign.

WORKED EXAMPLES · MISSION 04

EXAMPLE 1 · Combining Laws

Simplify: $x^5 \cdot x^3 / x^2$

1. **Top first.** Multiplying with same base — add powers.

   $x^5 \cdot x^3 = x^8$

2. **Now divide** — subtract powers.

   $x^8 / x^2 = \mathbf{x^6}$

EXAMPLE 2 · With Negative Exponents

Simplify (no negative exponents in final answer): $2x^{-3} \cdot 4x^5$

1. **Multiply coefficients** separately from variables.

   $(2 \cdot 4) \cdot (x^{-3} \cdot x^5) = 8 \cdot x^{-3+5} = 8x^2$

   $\mathbf{= 8x^2}$

   The negative exponent cancelled out cleanly. If a negative power remains, flip it to a denominator at the end.

EXAMPLE 3 · Power of a Quotient

Simplify: $(2x^2y^3)^3 / (4xy)^2$

1. **Apply the outer power to every factor** — top and bottom.

   $(2x^2y^3)^3 = 2^3 \cdot x^6 \cdot y^9 = 8x^6y^9$

   $(4xy)^2 = 16x^2y^2$

2. **Divide.**

   $8x^6y^9 / 16x^2y^2 = (8/16) \cdot x^{6-2} \cdot y^{9-2}$

   $\mathbf{= (1/2)x^4y^7}$ or $x^4y^7 / 2$

EXAMPLE 4 · Exponential Equation (Same Base)

Solve: $3^{x+1} = 81$

1. **Rewrite the right side** with the same base.

   $81 = 3^4$, so $3^{x+1} = 3^4$

2. **Same base, so the exponents must be equal.**

   $x + 1 = 4$

   $\mathbf{x = 3}$

   Golden rule: get the same base on both sides, then equate exponents.

PRO TIP — MEMORISE THESE POWERS COLD

Know by heart: powers of 2 (2, 4, 8, 16, 32, 64, 128, 256, 512, 1024), powers of 3 (3, 9, 27, 81, 243, 729), powers of 5 (5, 25, 125, 625), and squares up to 15^2. They're the building blocks of every exponential equation in Grade 10–12.

DRILLS · MISSION 04 · SIMPLIFY OR SOLVE

1. $x^4 \cdot x^5$
2. $x^{10} \div x^3$
3. $(x^3)^4$
4. $(2x^2)^3$
5. $x^0 + 5x^0$
6. $3x^{-2}$ (no negative exponents)
7. $(4x^3y) \cdot (2x^{-1}y^4)$
8. $2^x = 32 \rightarrow$ find x

PRO TIP — BRACKET-EVERYTHING STRATEGY

When a power is outside a bracket, write out the distribution explicitly: $(2x^3)^4 = 2^4 \cdot (x^3)^4 = 16x^{12}$. Don't try to do it all in your head — that's where the marks vanish.

⚔ BOSS BATTLE 04 · THE POWER WARDEN

Exam-style. Time yourself: 12 minutes. Solutions on p. 46.

1. Simplify (no negative exponents): $(3x^2y^{-3})^2 \cdot (xy^2)^3$
2. Simplify: $(2x^3)^2 \cdot 4x^{-1} / 8x^4$
3. Solve for x: $5^{2x-1} = 125$
4. Solve for x: $2^{x+3} = 4^{x-1}$ [rewrite 4 as 2^2]
5. Simplify: $(a^3b^{-2}) / (a^{-1}b^3)$ [combine carefully]

MISSION 04 SELF-CHECK

☐ Still confused ☐ Getting there ☐ Solid

GAME LINK

Open Algebra Assault → Mission 04: The Power Warden. The Warden shifts between bases (2, 3, 5, 10). Match the base before you fire. Same base = open shield.

MISSION 05

Inequalities

THE SIGN-FLIP · Boss: The Sign-Flipper

The Hook

An inequality is just an equation with a $<$, $>$, $\le$, or $\ge$ instead of $=$. You solve it almost exactly the same way as a linear equation — with **one critical difference**.

The Symbols

Symbol	Means	Includes the boundary?
$<$	less than	No
$>$	greater than	No
$\le$	less than or equal to	Yes (closed dot)
$\ge$	greater than or equal to	Yes (closed dot)

THE GOLDEN RULE — THE SIGN FLIP

When you **multiply or divide** both sides of an inequality by a **negative number**, you MUST flip the inequality sign.

$$\text{If} \quad -2x > 6 \quad \text{then dividing by } -2 \text{ gives} \quad x < -3$$

Note: $>$ flipped to $<$. That's the rule. Forget it, and the entire question is wrong.

TRAP #1 — FORGETTING TO FLIP

By far the most common Grade 10 inequality error. Marker is *waiting* for it.

TRAP #2 — FLIPPING WHEN YOU SHOULDN'T

Adding or subtracting a negative number does NOT flip the sign. Only multiplying or dividing by negative. $x - 5 > 3 \rightarrow x > 8$ (no flip; you added 5).

Showing the Answer on a Number Line

Inequality answers are **ranges**, not single points. Represent them on a number line:

- **Open circle** ○ for $<$ or $>$ (boundary not included)
- **Closed circle** ● for $\le$ or $\ge$ (boundary included)
- Arrow points in the direction the variable goes

e.g. $x > 3$ → —○══►

e.g. $x \le -1$ → ◄══●—

WORKED EXAMPLES · MISSION 05

EXAMPLE 1 · Basic Inequality (No Flip)

Solve: $2x - 5 < 11$

1. **Add 5** to both sides.

   $2x < 16$

2. **Divide both sides by 2.** 2 is positive — no flip.

   $\mathbf{x < 8}$

   Number line: open circle at 8, arrow pointing left.

EXAMPLE 2 · THE SIGN FLIP

Solve: $-3x + 4 > 10$

1. **Subtract 4** from both sides. (No flip — subtraction.)

   $-3x > 6$

2. **Divide both sides by −3.** ⚠ FLIP THE SIGN.

   $\mathbf{x < -2}$

   Number line: open circle at −2, arrow pointing left. If you forgot to flip you'd have written x > −2 — wrong direction, half the marks gone.

EXAMPLE 3 · Compound (Sandwich) Inequality

Solve: $-2 \le 3x - 1 < 8$

1. **Apply the same operation to all three parts.** Add 1 everywhere.

   $-1 \le 3x < 9$

2. **Divide everywhere by 3.** Positive — no flip.

   $\mathbf{-1/3 \le x < 3}$

   Number line: closed circle at −1/3, open circle at 3, line between them.

EXAMPLE 4 · Inequality Word Problem

Question: A taxi charges a R15 base fare plus R8 per kilometre. How far can you travel for at most R75?

1. **Define.** Let k = number of kilometres.
2. **Set up.** "At most R75" means ≤ 75.

$$15 + 8k \leq 75$$

3. **Solve.**

$$8k \leq 60 \quad \rightarrow \quad k \leq 7.5$$

You can travel up to 7.5 km.

Real-world check: at exactly 7.5 km, fare = 15 + 8(7.5) = 75 ✓

PRO TIP — SUBSTITUTE A TEST VALUE

Got $x < 4$? Pick any value less than 4 (say $x = 0$) and plug it into the original inequality. If it makes the original statement true, you've got the direction right. If false, you flipped the wrong way.

DRILLS · MISSION 05 · SOLVE AND SHOW ON NUMBER LINE

1. $x + 4 > 9$
2. $3x - 2 \leq 13$
3. $-x < 5$
4. $-2x \geq 10$
5. $4 - 3x > 1$
6. $5(x - 1) < 2x + 4$
7. $-4 \leq 2x - 6 < 8$
8. $(x + 2)/3 \geq x - 4$

PRO TIP — WRITE THE FLIPPED SIGN CLEARLY

When you flip, draw a small arrow above the new sign so the marker sees you knew to flip it. Examiners give method marks for showing the rule was applied.

⚔ BOSS BATTLE 05 · THE SIGN-FLIPPER

Exam-style. Time yourself: 12 minutes. Solutions on p. 47.

1. Solve and represent on a number line: $7 - 2x \leq 13$
2. Solve: $3(x - 2) > 5(x + 1) - 9$
3. Solve compound: $-5 < 2x + 1 \leq 7$
4. The temperature in Cape Town today must be at least 12°C and below 28°C. Write this as a compound inequality with T = temperature.
5. Solve: $(2x - 1)/3 - (x + 2)/2 \leq 1$ [clear fractions first; watch the inequality direction]

MISSION 05 SELF-CHECK

☐ Still confused ☐ Getting there ☐ Solid

GAME LINK

Open Algebra Assault → Mission 05: The Sign-Flipper. The Flipper attacks with negative coefficients — every successful hit means you remembered to flip. Miss the flip and your shot reverses on you.

FINAL BOSS

Mixed Mission

ALL FIVE WEAPONS · The Algebra Citadel

Real exams don't tell you which weapon to use. You have to read the question and decide. Below: 10 mixed questions spanning all five missions, in random order. Time yourself: **30 minutes**. Full solutions on p. 47.

⚔ FINAL BOSS · THE ALGEBRA CITADEL · 30 MIN

1. Solve for x: $4(x - 3) - 2(x + 1) = x - 5$
2. Solve: $x^2 - 4x - 21 = 0$
3. Solve simultaneously: $2x - 3y = 7$; $x + 2y = 0$
4. Simplify (no negative exponents): $(5x^{-2}y^3)^2 \cdot (2xy)^{-1}$
5. Solve: $3 - 2x > 5(x - 4)$
6. Solve for x: $2^{x+2} = 32$
7. The length of a rectangle is 3 cm more than its width. The area is 40 cm^2. Find the dimensions.
8. Solve: $-1 \leq (1 - 2x)/3 < 5$
9. Two numbers differ by 4. Their sum is 22. Find them. [simultaneous]
10. Solve for x: $(x - 1)(x + 3) = 12$ ⚠ standard form!

FINAL BOSS — HOW DID THAT GO?

☐ Hurt me ☐ Partial credit ☐ Crushed it

SCORING YOURSELF

8–10 correct: You're in great shape for the Grade 10 test. 5–7 correct: You're solid on most weapons but one or two need a re-run. 0–4 correct: This is normal at first. Pick your weakest mission and re-do it. No shame — that's how every strong learner gets strong.

ARMOURY · FULL SOLUTIONS

Boss Battle Solutions

Boss Battle 01 — The Slope Sentinel

1. $5(2x - 3) - 4(x - 1) = 9 \to 10x - 15 - 4x + 4 = 9 \to 6x - 11 = 9 \to 6x = 20 \to$ **$x = 10/3$**
2. $(x - 2)/3 - (x + 1)/4 = 1$. Multiply by 12: $4(x - 2) - 3(x + 1) = 12 \to 4x - 8 - 3x - 3 = 12 \to x - 11 = 12 \to$ **$x = 23$**
3. Let width = w. Length = w + 5. Perimeter = $2(w + w + 5) = 4w + 10 = 46 \to 4w = 36 \to w = 9$. **Width = 9 cm, length = 14 cm.**
4. $s = ut + (1/2)at^2$. Subtract $(1/2)at^2$: $s - (1/2)at^2 = ut$. Divide by t: **$u = (s - (1/2)at^2) / t$** or equivalently $u = s/t - (1/2)at$.
5. $3(x - 2) - 2(2x + 1) = -x - 8 \to 3x - 6 - 4x - 2 = -x - 8 \to -x - 8 = -x - 8 \to 0 = 0$. **All real x** (identity — every value of x works).

Boss Battle 02 — The Parabola Beast

1. $x^2 - 11x + 28 = 0 \to (x - 7)(x - 4) = 0 \to$ **$x = 7$ or $x = 4$**
2. $2x^2 - 5x - 3 = 0 \to (2x + 1)(x - 3) = 0 \to$ **$x = -1/2$ or $x = 3$**
3. $x(x - 4) = 5(x - 4) \to x^2 - 4x = 5x - 20 \to x^2 - 9x + 20 = 0 \to (x - 4)(x - 5) = 0 \to$ **$x = 4$ or $x = 5$**
4. Let smaller = x. Then $x(x + 1) = 72 \to x^2 + x - 72 = 0 \to (x + 9)(x - 8) = 0 \to x = -9$ (reject, must be positive) or $x = 8$. **Integers are 8 and 9.**
5. $(2x + 1)^2 = 25 \to 2x + 1 = \pm 5$. Case A: $2x + 1 = 5 \to x = 2$. Case B: $2x + 1 = -5 \to x = -3$. **$x = 2$ or $x = -3$.**

Boss Battle 03 — The Twin Tyrants

1. $2x + 5y = 16$ (1); $3x - 2y = 5$ (2). Multiply (1) by 2 and (2) by 5: $4x + 10y = 32$; $15x - 10y = 25$. Add: $19x = 57 \rightarrow x = 3$. Sub into (1): $6 + 5y = 16 \rightarrow y = 2$. **$x = 3, y = 2$**
2. $x/2 + y/3 = 5$. Multiply by 6: $3x + 2y = 30$ (1). $x - y = 4 \rightarrow x = y + 4$ (2). Sub into (1): $3(y + 4) + 2y = 30 \rightarrow 5y + 12 = 30 \rightarrow 5y = 18 \rightarrow y = 3.6 = 18/5$. Then $x = 18/5 + 20/5 = 38/5 = 7.6$. **$x = 38/5, y = 18/5$**. (Exam-style answers may be left as fractions.)
3. $5o + 3a = 36$ (1); $2o + 4a = 34 \rightarrow o + 2a = 17 \rightarrow o = 17 - 2a$ (2). Sub: $5(17 - 2a) + 3a = 36 \rightarrow 85 - 10a + 3a = 36 \rightarrow -7a = -49 \rightarrow a = 7$. Then $o = 17 - 14 = 3$. **Orange R3, Apple R7.**
4. Tens = x, units = y. $x + y = 11$ (1). Original $10x + y$; reversed $10y + x$. $10x + y - (10y + x) = 27 \rightarrow 9x - 9y = 27 \rightarrow x - y = 3$ (2). Add (1)+(2): $2x = 14 \rightarrow x = 7, y = 4$. **Number is 74.**
5. Set $2x - 1 = -x + 5 \rightarrow 3x = 6 \rightarrow x = 2$; then $y = 3$. **Intersection: (2; 3).**

Boss Battle 04 — The Power Warden

1. $(3x^2y^{-3})^2 \cdot (xy^2)^3 = 9x^4y^{-6} \cdot x^3y^6 = 9x^7y^0 =$ **$9x^7$**
2. $(2x^3)^2 \cdot 4x^{-1} / 8x^4 = 4x^6 \cdot 4x^{-1} / 8x^4 = 16x^5 / 8x^4 =$ **$2x$**
3. $5^{2x - 1} = 125 = 5^3$. So $2x - 1 = 3 \rightarrow$ **$x = 2$**
4. $2^{x+3} = 4^{x - 1} = (2^2)^{x - 1} = 2^{2x - 2}$. So $x + 3 = 2x - 2 \rightarrow$ **$x = 5$**
5. $(a^3b^{-2}) / (a^{-1}b^3) = a^{3 - (-1)} \cdot b^{-2 - 3} = a^4b^{-5} =$ **a^4 / b^5**

Boss Battle 05 — The Sign-Flipper

1. $7 - 2x \leq 13 \rightarrow -2x \leq 6 \rightarrow$ divide by -2 and FLIP: **$x \geq -3$**. Number line: closed circle at -3, arrow right.
2. $3(x - 2) > 5(x + 1) - 9 \rightarrow 3x - 6 > 5x + 5 - 9 \rightarrow 3x - 6 > 5x - 4 \rightarrow -2x > 2 \rightarrow$ divide by -2 and FLIP: **$x < -1$**
3. $-5 < 2x + 1 \leq 7 \rightarrow$ subtract 1: $-6 < 2x \leq 6 \rightarrow$ divide by 2: **$-3 < x \leq 3$**
4. **$12 \leq T < 28$**
5. $(2x - 1)/3 - (x + 2)/2 \leq 1$. Multiply by 6: $2(2x - 1) - 3(x + 2) \leq 6 \rightarrow 4x - 2 - 3x - 6 \leq 6 \rightarrow x - 8 \leq 6 \rightarrow$ **$x \leq 14$**. (Multiplied by positive 6, no flip.)

Final Boss — The Algebra Citadel

1. $4(x - 3) - 2(x + 1) = x - 5 \rightarrow 4x - 12 - 2x - 2 = x - 5 \rightarrow 2x - 14 = x - 5 \rightarrow$ **$x = 9$**
2. $x^2 - 4x - 21 = 0 \rightarrow (x - 7)(x + 3) = 0 \rightarrow$ **$x = 7$ or $x = -3$**
3. $2x - 3y = 7$ (1); $x + 2y = 0 \rightarrow x = -2y$ (2). Sub: $2(-2y) - 3y = 7 \rightarrow -7y = 7 \rightarrow y = -1$. Then $x = 2$. **$x = 2, y = -1$**
4. $(5x^{-2}y^3)^2 \cdot (2xy)^{-1} = 25x^{-4}y^6 \cdot (1/(2xy)) = 25y^6 / (2xy \cdot x^4) =$ **$25y^5 / (2x^5)$**
5. $3 - 2x > 5x - 20 \rightarrow 3 + 20 > 5x + 2x \rightarrow 23 > 7x \rightarrow$ **$x < 23/7$** (no flip — divided by positive 7)
6. $2^{x + 2} = 32 = 2^5$. So $x + 2 = 5 \rightarrow$ **$x = 3$**
7. Let width = w, length = w + 3. Area: $w(w + 3) = 40 \rightarrow w^2 + 3w - 40 = 0 \rightarrow (w + 8)(w - 5) = 0 \rightarrow w = 5$ (reject −8, width must be positive). **Width = 5 cm, length = 8 cm.**
8. $-1 \leq (1 - 2x)/3 < 5 \rightarrow$ multiply by 3 (positive, no flip): $-3 \leq 1 - 2x < 15 \rightarrow$ subtract 1: $-4 \leq -2x < 14 \rightarrow$ divide by −2 and FLIP all signs: $2 \geq x > -7 \rightarrow$ **$-7 < x \leq 2$**
9. Let numbers = x and y, x > y. $x - y = 4$ (1); $x + y = 22$ (2). Add: $2x = 26 \rightarrow x = 13, y = 9$. **Numbers are 13 and 9.**
10. $(x - 1)(x + 3) = 12 \rightarrow x^2 + 2x - 3 = 12 \rightarrow x^2 + 2x - 15 = 0 \rightarrow (x + 5)(x - 3) = 0 \rightarrow$ **$x = -5$ or $x = 3$**

DEBRIEF — WHERE DID YOU LOSE MARKS?

Look back at any questions you got wrong. For each, write ONE sentence: "*I lost this because...*". Patterns reveal themselves fast. If three of your mistakes were the same kind (sign errors, factorisation, exponent law confusion), that's your single most important weakness — and that's the chapter to re-run before your next test.

BASE CAMP · WHAT'S NEXT

Mission complete.
Now what?

> You now own the Grade 10 algebra core for CAPS, IEB, and Cambridge IGCSE. The next move is reps — and a real human if you need one.

Your next 7 days

- **Day 1–2:** Replay every Algebra Assault mission until you can clear each with a 5-streak.
- **Day 3–4:** Redo any Boss Battle you scored under 3/5 on. Don't peek at solutions.
- **Day 5:** Find one past paper (CAPS / IEB / Cambridge) and try Section A only.
- **Day 6:** Re-attempt the Final Boss under exam conditions.
- **Day 7:** Rest. Sleep is a maths skill.

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